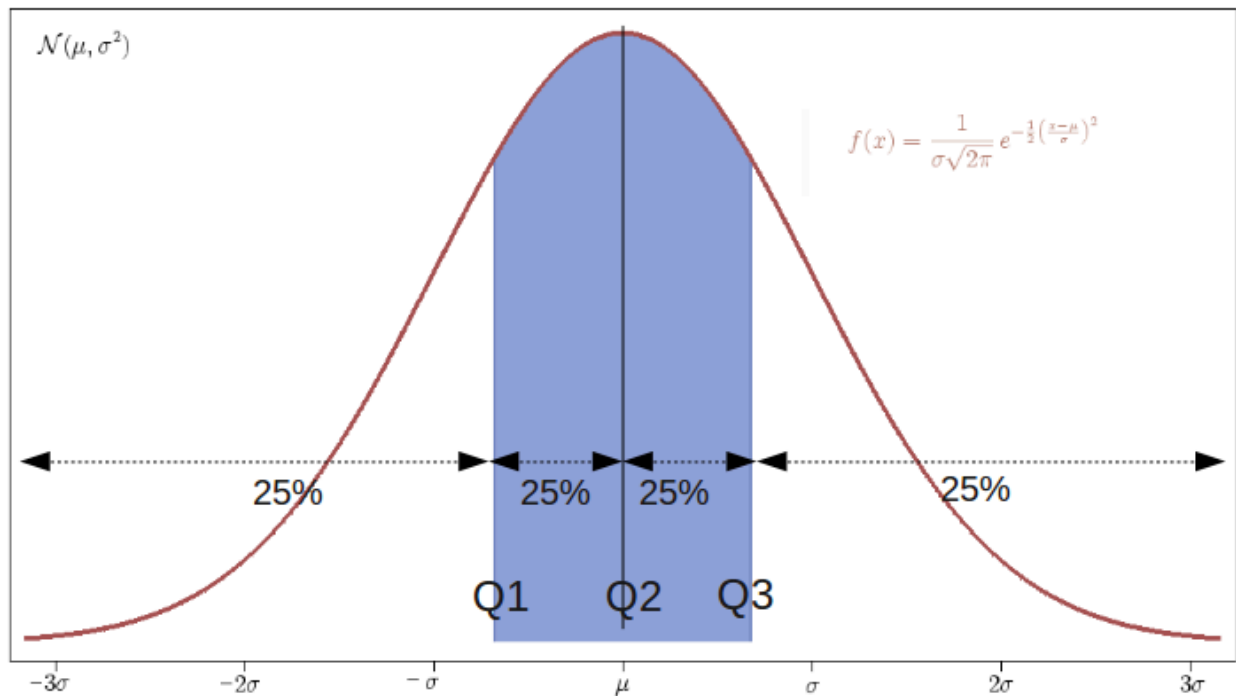


Quantiles



Quantiles determine the value below which a certain percentage of data falls

- median = 0.5 quantile = the middle value that divide the data
- Quartiles = divide the data into four part: 25th percentile ($Q1$), 50th percentile ($Q2$ /median), and 75th percentile ($Q3$)
- Percentile = divide the data into 100 equal parts
- deciles = divide the data into 10 equal parts

Calculation process

1. **Order the Data:** Sort the dataset from smallest to largest.
2. **Calculate Position:** Determine the position of the desired quantile, for example, using the formula $p(n + 1)$ where p is the percentile and n is the sample size
3. **Interpolate:** If the quantile lies between two data points, interpolate between them.